

Dynamic Replacement Model: Fit Summary across 4p, 8p, and 8p+FE Specifications

Kaleb Javier

2026-05-18

Table of contents

1	Introduction	1
2	Parameter Estimates	2
2.1	4-parameter specification	2
2.2	8-parameter specification	2
2.3	8-parameter + state fixed effects	2
2.3.1	State fixed-effect coefficients	4
3	Model Comparison	4
4	Goodness of Fit: Model-Implied vs. Empirical Cell Shares	5
4.1	4-parameter fit	5
4.2	8-parameter fit	6
4.3	8-parameter + FE fit	8
5	Appendix: Per-cell fit numbers	8
5.1	4-parameter	8
5.2	8-parameter	10
5.3	8-parameter + FE	10

1 Introduction

This report summarizes the in-sample fit of the dynamic structural replacement model under three nested specifications, each estimated by Aguirregabiria-Mira NPL on the observed UST panel ($N = 2,282,735$ tank-year observations):

1. **4-parameter** — a single $(\kappa, K, \gamma_{\text{price}}, \gamma_{\text{risk}})$ block, pooled across wall and regime.
2. **8-parameter** — the same primitives split by wall type (SW/DW) and insurance regime (FF/RB): 4 cost parameters and 4 price/risk slopes.
3. **8-parameter + state fixed effects** — the 8p structural specification augmented with 17 state-level intercepts α_g that enter the measurement equation (not the agent's equilibrium) for the 17 control states; Texas is the baseline ($\alpha_{\text{TX}} \equiv 0$).

Table 1: 4-parameter NPL estimates: pooled κ , K , γ_{price} , γ_{risk} .

	Observed (headline) TX 2006+, controls 1999+	Extended (2000+) TX 2000+, controls 2000+
κ_{exit}	\$261,222 (\$306) [\$260,621, \$261,822]	\$257,702 (\$310) [\$257,094, \$258,310]
K	\$51,779 (\$142) [\$51,501, \$52,057]	\$52,658 (\$146) [\$52,371, \$52,945]
γ_{price}	-0.293 (0.049) [-0.389, -0.197]	-1.142 (0.048) [-1.236, -1.048]
γ_{risk}	0.164 (0.003) [0.158, 0.170]	0.153 (0.003) [0.147, 0.159]
$\log L$	-268,481	-247,614
N obs	2,282,735	2,300,780

Point estimates with AM (2002) profile-likelihood standard errors in parentheses and 95% CIs in brackets. Bootstrap SEs deferred.

The report is organized for direct visual and statistical comparison: Section 2 reports the three parameter tables and the FE alphas; Section 3 reports the three-way information-criteria and likelihood-ratio table; Section 4 reports model-implied vs. empirical action shares at the (wall $\times$ regime $\times$ age bin) cell level for each specification, with figures produced by a single shared plotting routine to guarantee structural parity. Per-cell numerical fit tables are in Section 5.

All artifacts are produced by `Code/Dynamic_Model/04k_Fit_Report_Artifacts.R` from saved fit objects under `Output/Estimation_Results/`. No estimation is re-run in the production of this report.

2 Parameter Estimates

2.1 4-parameter specification

Table 1 reports point estimates with AM (2002) profile-likelihood standard errors and 95% Wald confidence intervals.

2.2 8-parameter specification

Table 2 splits κ and K by wall type (SW/DW) and the price/risk slopes by insurance regime (FF/RB).

2.3 8-parameter + state fixed effects

Table 3 reports the 8-structural-parameter block from the joint (θ, α) estimation. The 17 state fixed effects are reported separately in Table 4.

Table 2: 8-parameter NPL estimates: κ and K split by wall (SW/DW); γ_{price} and γ_{risk} split by regime (FF/RB).

	Observed (headline) TX 2006+, controls 1999+
κ_{SW}	\$236,795 (\$716) [\$235,392, \$238,198]
κ_{DW}	\$211,775 (\$856) [\$210,098, \$213,452]
K_{SW}	\$58,085 (\$212) [\$57,670, \$58,501]
K_{DW}	\$43,301 (\$244) [\$42,823, \$43,779]
$\gamma_{\text{price,FF}}$	-14.382 (0.293) [-14.957, -13.808]
$\gamma_{\text{price,RB}}$	-7.485 (0.151) [-7.781, -7.188]
$\gamma_{\text{risk,FF}}$	0.065 (0.005) [0.055, 0.075]
$\gamma_{\text{risk,RB}}$	-0.083 (0.009) [-0.100, -0.066]
$\log L$	-261,770
N obs	2,282,735

Point estimates with AM (2002) profile-likelihood SEs in parentheses and 95% CIs in brackets. MC robustness battery on 8-param spec deferred.

Table 3: 8-parameter + FE NPL estimates: structural parameters from the joint (θ, α) fit. The 17 state fixed effects α_g are reported separately in Table 4 (omitted here to avoid duplication).

Parameter	Estimate	(SE)	95% CI
κ_{SW}	\$246,706	(\$811)	[\$245,118, \$248,295]
κ_{DW}	\$219,264	(\$925)	[\$217,451, \$221,078]
K_{SW}	\$55,331	(\$362)	[\$54,622, \$56,041]
K_{DW}	\$43,051	(\$359)	[\$42,347, \$43,756]
$\gamma_{\text{price,FF}}$	-11.088	(0.343)	[-11.760, -10.416]
$\gamma_{\text{price,RB}}$	-6.506	(0.163)	[-6.826, -6.185]
$\gamma_{\text{risk,FF}}$	0.088	(0.005)	[0.078, 0.098]
$\gamma_{\text{risk,RB}}$	-0.120	(0.009)	[-0.137, -0.102]
$\log L$	-251,613		
N obs	2,282,735		

Observed sample. TX FE fixed at 0. 17 control-state maintain-shifters estimated.

Table 4: State fixed-effect coefficients $\hat{\alpha}_g$ from the 8p+FE specification. Texas is the baseline ($\alpha_{TX} \equiv 0$); standard errors in parentheses.

State	$\hat{\alpha}_g$ (SE)
TX	0 (baseline)
AR	0.095 (0.040)
CO	0.064 (0.042)
ID	-0.114 (0.049)
KS	0.478 (0.043)
KY	-0.124 (0.038)
LA	-0.449 (0.037)
MA	-0.415 (0.039)
MD	-0.360 (0.037)
ME	-1.393 (0.042)
MN	0.290 (0.040)
MO	8.181 (0.774)
NC	0.572 (0.038)
OH	0.064 (0.037)
OK	0.151 (0.039)
SD	2.269 (0.083)
TN	0.187 (0.038)
VA	0.803 (0.039)

Texas is the baseline (alpha set to 0). Other states are estimated jointly with structural theta in the 25-parameter 8p+FE specification.

2.3.1 State fixed-effect coefficients

The state fixed effects α_g enter the measurement equation (the likelihood) only — they shift the model-implied probability of *Maintain* in state g relative to the TX baseline. They do **not** enter the agent’s equilibrium value function during counterfactual re-solves, per the “Semantic 2” convention adopted in this codebase.

Note on extreme estimates. The Missouri ($\hat{\alpha}_{MO} = 8.18$, SE 0.77) and South Dakota ($\hat{\alpha}_{SD} = 2.27$, SE 0.08) coefficients are substantially larger than the other 15 control-state alphas, which span roughly $[-1.4, +0.8]$. Both are statistically distinguishable from zero and reflect near-zero observed *Exit* and *Replace* activity in those state-year cells relative to the TX baseline; they are not numerical artifacts.

3 Model Comparison

Table 5 reports log-likelihood, information criteria, and nested likelihood-ratio tests across the three specifications. All three are fit on the same observed sample, so the comparison is direct.

Both nested restrictions are rejected at any conventional level. AIC and BIC agree on the ranking $4p \succ 8p \succ 8p+FE$ (lower is better), with the BIC penalty large enough to make 8p+FE still preferred despite the 17 additional parameters.

Table 5: Three-way nested model comparison. $AIC = -2\log L + 2k$; $BIC = -2\log L + k\log N$. Likelihood-ratio tests: 8p vs 4p ($\Delta k = 4$), 8p+FE vs 8p ($\Delta k = 17$). Both LR statistics are far in the tail of their reference χ^2 distributions, with p -values numerically zero.

Model	k	N	$\log L$	AIC	BIC	LR vs nested	df	p -value
4-param	4	2,282,735	-268,481	536,971	537,021	—	—	—
8-param	8	2,282,735	-261,770	523,557	523,658	13,422	4	0
8-param+FE	25	2,282,735	-251,613	503,277	503,593	20,314	17	0

LR tests: 8p vs 4p, then 8p+FE vs 8p.

4 Goodness of Fit: Model-Implied vs. Empirical Cell Shares

This section reports the model-implied conditional choice probability $\hat{P}(\text{action} \mid \text{state})$ at the converged $\hat{\theta}$ alongside the empirical action share in each (wall $\times$ regime $\times$ age bin) cell. Empirical shares are computed by direct cell aggregation on the observed panel. Model-implied shares are read from the NPL-converged $\hat{P}$ matrix saved in each fit object. For the 8p+FE specification, $\hat{P}$ is the alpha-augmented, g -marginalized measurement probability (the same object the NPL outer loop uses for its CCP update).

Each model has two figures (one per wall type), each with three facets (Maintain / Exit / Replace) and two colored lines (FF / RB regime). All six figures are produced by a single shared plotting routine so visual comparability across specifications is structural.

4.1 4-parameter fit

Figure 1: 4-parameter fit, Single-Walled tanks. Lines are model-implied $\hat{P}(\text{action} \mid \text{state})$; open circles are empirical cell shares with area proportional to cell sample size.

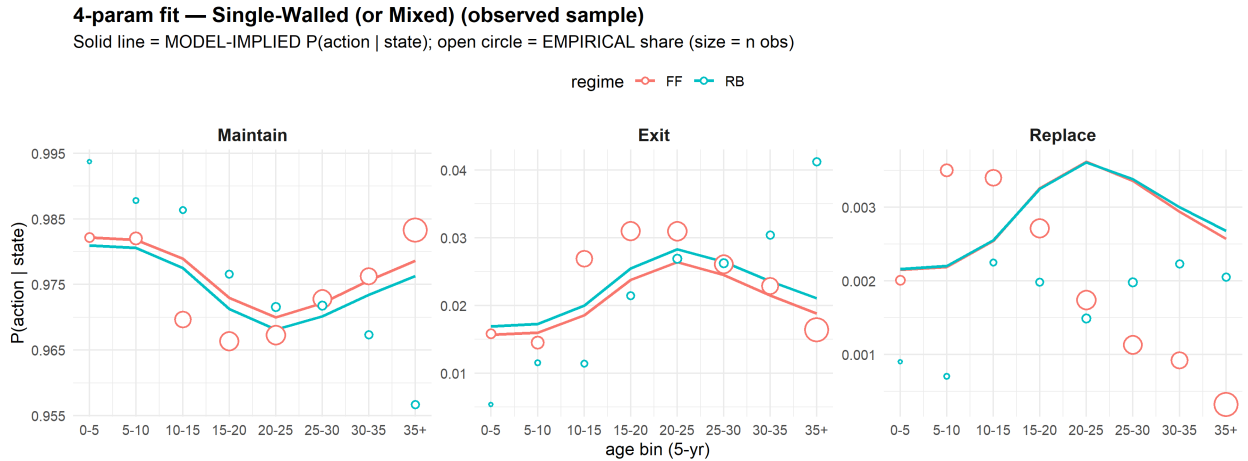
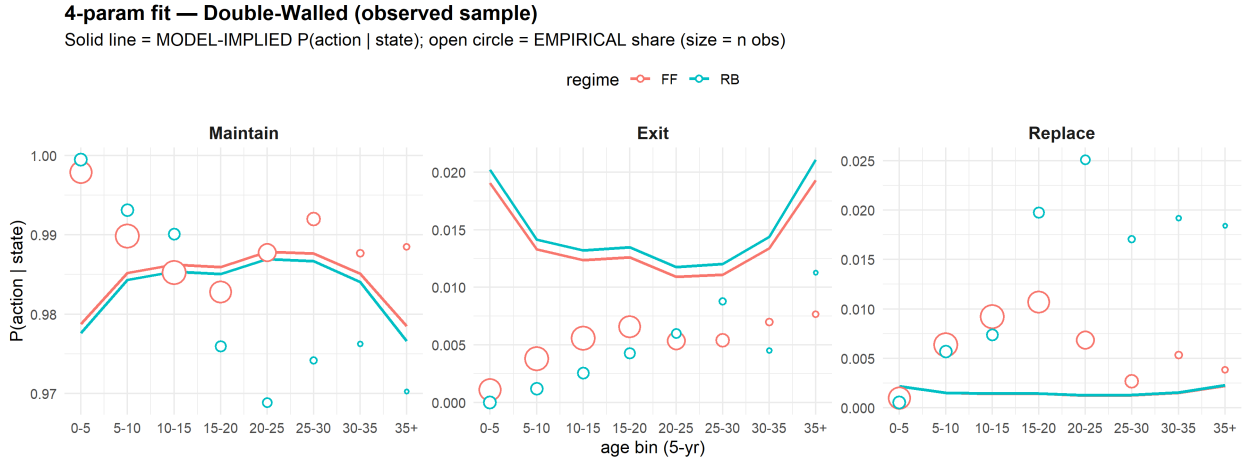


Table 6: 4-parameter weighted-RMSE summary of model-vs-empirical residuals by wall, regime, and action. Residual = empirical – model share; weights are cell sample size.

wall	regime	action	N_cells	total_n	unweighted_mean_residual	weighted_mean_residual	unweighted_RMSE	weighted_RMSE	max_abs_residual
DW	FF	Exit	8	351597	-0.00870	-0.00913	0.00959	0.01014	0.01795
DW	RB	Exit	8	78616	-0.01022	-0.01222	0.01125	0.01326	0.02022
SW	FF	Exit	8	1656628	0.00242	0.00237	0.00441	0.00453	0.00839
SW	RB	Exit	8	195894	-0.00055	0.00212	0.00943	0.00923	0.02020
DW	FF	Maintain	8	351597	0.00456	0.00411	0.00809	0.00872	0.01913
DW	RB	Maintain	8	78616	-0.00231	0.00351	0.01247	0.01396	0.02186
SW	FF	Maintain	8	1656628	-0.00156	-0.00106	0.00446	0.00472	0.00924
SW	RB	Maintain	8	195894	0.00170	-0.00093	0.00972	0.00926	0.01957
DW	FF	Replace	8	351597	0.00414	0.00502	0.00526	0.00616	0.00923
DW	RB	Replace	8	78616	0.01253	0.00871	0.01493	0.01238	0.02386
SW	FF	Replace	8	1656628	-0.00086	-0.00131	0.00160	0.00179	0.00225
SW	RB	Replace	8	195894	-0.00116	-0.00119	0.00127	0.00132	0.00212

Figure 2: 4-parameter fit, Double-Walled tanks. Same construction as Figure 1.



4.2 8-parameter fit

Figure 3: 8-parameter fit, Single-Walled tanks. Same construction as Figure 1.

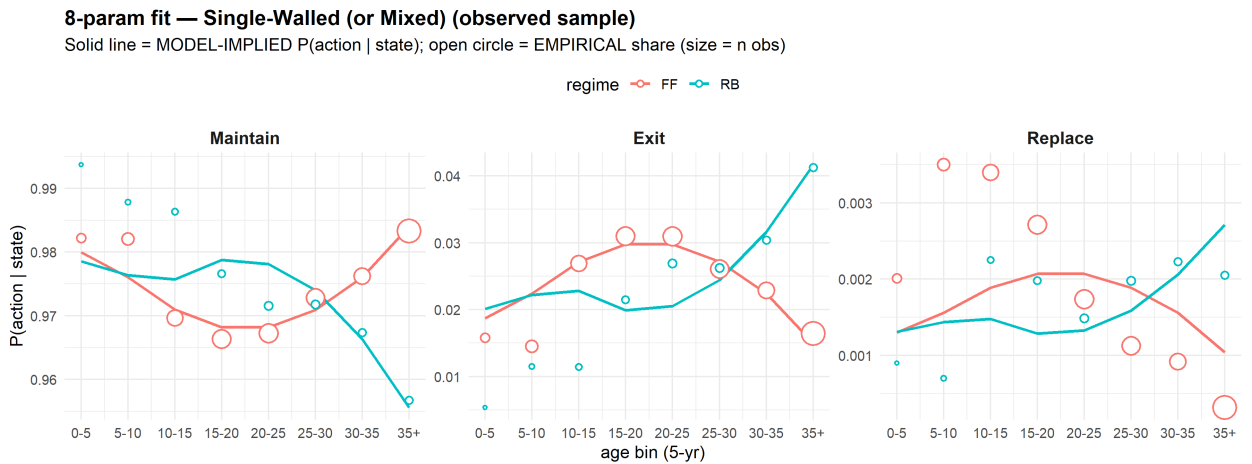
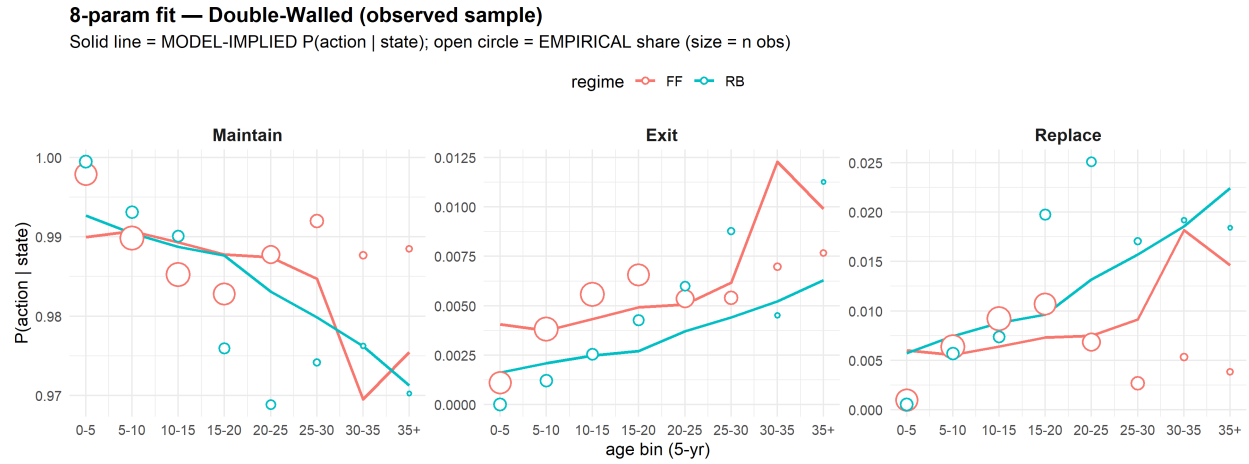


Table 7: 8-parameter weighted-RMSE summary by wall, regime, and action. Same construction as Table 6.

wall	regime	action	N_cells	total_n	weighted_RMSE	weighted_mean_residual	max_abs_residual
DW	FF	Exit	8	351597	0.00170	-0.00006	0.00532
DW	RB	Exit	8	78616	0.00176	0.00024	0.00497
SW	FF	Exit	8	1656628	0.00227	0.00005	0.00792
SW	RB	Exit	8	195894	0.00558	-0.00042	0.01475
DW	FF	Maintain	8	351597	0.00537	0.00031	0.01818
DW	RB	Maintain	8	78616	0.00773	-0.00140	0.01421
SW	FF	Maintain	8	1656628	0.00193	-0.00003	0.00598
SW	RB	Maintain	8	195894	0.00563	0.00027	0.01516
DW	FF	Replace	8	351597	0.00378	-0.00024	0.01286
DW	RB	Replace	8	78616	0.00637	0.00117	0.01192
SW	FF	Replace	8	1656628	0.00090	-0.00002	0.00194
SW	RB	Replace	8	195894	0.00051	0.00015	0.00077

Figure 4: 8-parameter fit, Double-Walled tanks. Same construction as Figure 1.



4.3 8-parameter + FE fit

Figure 5: 8-parameter + FE fit, Single-Walled tanks. Model-implied probabilities marginalize the per- g choice probability over the empirical state composition $w_g(s)$. Same plot construction as Figure 1.

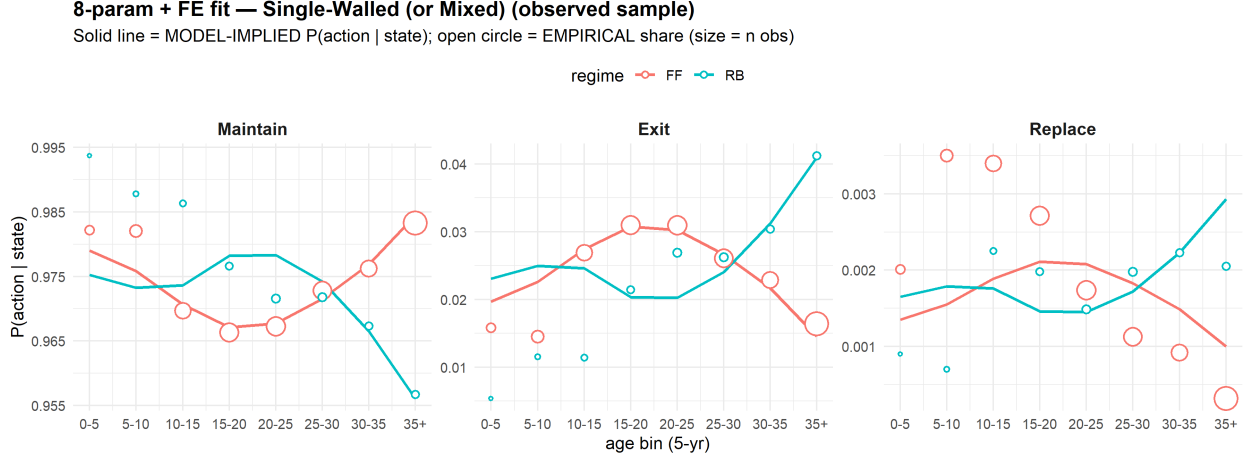
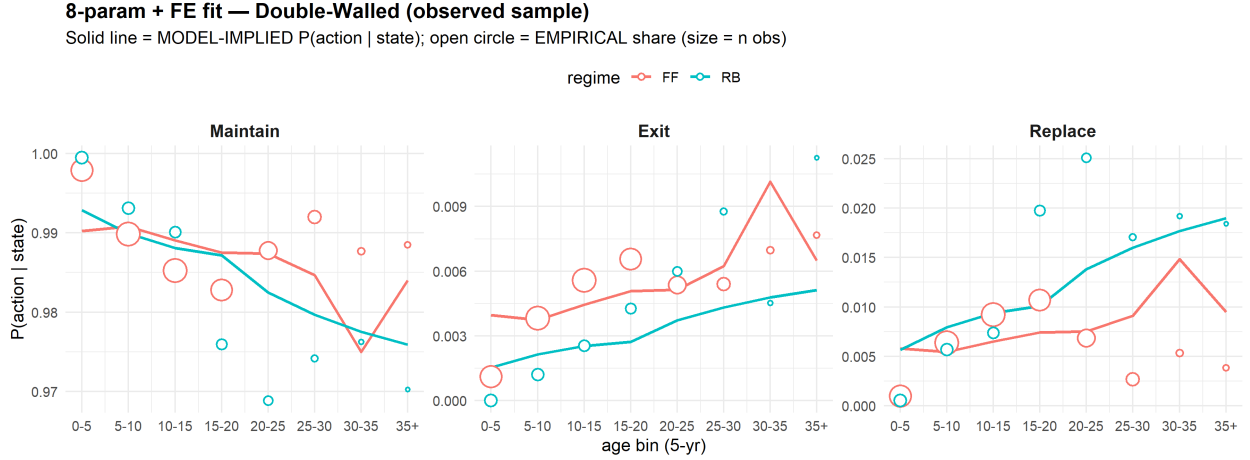


Figure 6: 8-parameter + FE fit, Double-Walled tanks. Same construction as Figure 5.



5 Appendix: Per-cell fit numbers

The following three tables report the full 32-row cell-level fit numbers for each specification. Columns: **Reg** (regime: FF/RB), **Wall** (SW/DW), **Age** (5-year bin), **N** (cell sample size), and for each of the three actions $a \in \{M, E, R\}$, **mod_a** = $\hat{P}_a(s)$ (model-implied), **emp_a** (empirical share), and **res_a** = $\text{emp_a} - \text{mod_a}$ (residual).

5.1 4-parameter

Table 8: 8-parameter + FE weighted-RMSE summary by wall, regime, and action. Same construction as Table 6.

wall	regime	action	N_cells	total_n	weighted_RMSE	weighted_mean_residual	max_abs_residual
DW	FF	Exit	8	351597	0.00154	-0.00005	0.00317
DW	RB	Exit	8	78616	0.00180	0.00026	0.00615
SW	FF	Exit	8	1656628	0.00235	0.00002	0.00810
SW	RB	Exit	8	195894	0.00642	-0.00054	0.01771
DW	FF	Maintain	8	351597	0.00486	0.00020	0.01267
DW	RB	Maintain	8	78616	0.00754	-0.00114	0.01359
SW	FF	Maintain	8	1656628	0.00184	-0.00003	0.00616
SW	RB	Maintain	8	195894	0.00652	0.00057	0.01846
DW	FF	Replace	8	351597	0.00345	-0.00015	0.00950
DW	RB	Replace	8	78616	0.00616	0.00088	0.01132
SW	FF	Replace	8	1656628	0.00088	0.00000	0.00195
SW	RB	Replace	8	195894	0.00052	-0.00003	0.00109

Table 9: 4-parameter per-cell fit: model-implied probability, empirical share, and residual by state cell.

Reg	Wall	Age	N	mod_M	emp_M	res_M	mod_E	emp_E	res_E	mod_R	emp_R	res_R
FF	DW	0-5	64621	0.9788	0.9979	0.0191	0.0191	0.0011	-0.0180	0.0022	0.0010	-0.0012
FF	DW	5-10	77103	0.9852	0.9898	0.0046	0.0133	0.0038	-0.0095	0.0015	0.0064	0.0048
FF	DW	10-15	77966	0.9862	0.9852	-0.0010	0.0124	0.0056	-0.0068	0.0014	0.0092	0.0078
FF	DW	15-20	62241	0.9860	0.9828	-0.0032	0.0126	0.0066	-0.0061	0.0014	0.0107	0.0092
FF	DW	20-25	41520	0.9878	0.9878	-0.0001	0.0109	0.0054	-0.0055	0.0012	0.0069	0.0056
FF	DW	25-30	20407	0.9876	0.9920	0.0043	0.0111	0.0054	-0.0057	0.0013	0.0026	0.0014
FF	DW	30-35	4872	0.9851	0.9877	0.0026	0.0134	0.0070	-0.0064	0.0015	0.0053	0.0038
FF	DW	35+	2867	0.9785	0.9885	0.0100	0.0193	0.0077	-0.0116	0.0022	0.0038	0.0016
FF	SW	0-5	46709	0.9822	0.9822	0.0000	0.0157	0.0158	0.0001	0.0022	0.0020	-0.0001
FF	SW	5-10	97999	0.9818	0.9820	0.0002	0.0160	0.0145	-0.0015	0.0022	0.0035	0.0013
FF	SW	10-15	180163	0.9789	0.9697	-0.0092	0.0185	0.0269	0.0084	0.0025	0.0034	0.0009
FF	SW	15-20	240344	0.9730	0.9663	-0.0066	0.0238	0.0310	0.0072	0.0033	0.0027	-0.0006
FF	SW	20-25	255278	0.9700	0.9673	-0.0027	0.0264	0.0310	0.0046	0.0036	0.0017	-0.0019
FF	SW	25-30	234030	0.9722	0.9728	0.0006	0.0245	0.0261	0.0016	0.0034	0.0011	-0.0022
FF	SW	30-35	191377	0.9756	0.9762	0.0006	0.0215	0.0228	0.0014	0.0029	0.0009	-0.0020
FF	SW	35+	410728	0.9786	0.9833	0.0046	0.0188	0.0164	-0.0024	0.0026	0.0003	-0.0022
RB	DW	0-5	18813	0.9776	0.9995	0.0219	0.0202	0.0000	-0.0202	0.0022	0.0005	-0.0016
RB	DW	5-10	16557	0.9843	0.9931	0.0088	0.0142	0.0012	-0.0130	0.0015	0.0057	0.0042
RB	DW	10-15	14499	0.9854	0.9901	0.0047	0.0132	0.0026	-0.0106	0.0014	0.0074	0.0060
RB	DW	15-20	13106	0.9851	0.9760	-0.0091	0.0135	0.0043	-0.0092	0.0014	0.0198	0.0183
RB	DW	20-25	8677	0.9870	0.9689	-0.0181	0.0118	0.0060	-0.0058	0.0013	0.0251	0.0239
RB	DW	25-30	4218	0.9867	0.9742	-0.0125	0.0120	0.0088	-0.0033	0.0013	0.0171	0.0158
RB	DW	30-35	1770	0.9841	0.9763	-0.0078	0.0144	0.0045	-0.0099	0.0015	0.0192	0.0177
RB	DW	35+	976	0.9766	0.9703	-0.0063	0.0211	0.0113	-0.0098	0.0023	0.0184	0.0162
RB	SW	0-5	3350	0.9810	0.9937	0.0128	0.0169	0.0054	-0.0115	0.0022	0.0009	-0.0013
RB	SW	5-10	9999	0.9806	0.9878	0.0072	0.0172	0.0115	-0.0057	0.0022	0.0007	-0.0015
RB	SW	10-15	19093	0.9775	0.9863	0.0088	0.0200	0.0114	-0.0085	0.0026	0.0022	-0.0003
RB	SW	15-20	26212	0.9713	0.9766	0.0053	0.0255	0.0214	-0.0040	0.0032	0.0020	-0.0013
RB	SW	20-25	36841	0.9681	0.9716	0.0035	0.0283	0.0269	-0.0014	0.0036	0.0015	-0.0021
RB	SW	25-30	37789	0.9702	0.9718	0.0016	0.0265	0.0262	-0.0003	0.0034	0.0020	-0.0014

(continued)

Reg	Wall	Age	N	mod_M	emp_M	res_M	mod_E	emp_E	res_E	mod_R	emp_R	res_R
RB	SW	30-35	31888	0.9734	0.9674	-0.0061	0.0236	0.0304	0.0069	0.0030	0.0022	-0.0008
RB	SW	35+	30722	0.9763	0.9567	-0.0196	0.0210	0.0412	0.0202	0.0027	0.0021	-0.0006

5.2 8-parameter

Table 10: 8-parameter per-cell fit. Same construction as Table 9.

Reg	Wall	Age	N	mod_M	emp_M	res_M	mod_E	emp_E	res_E	mod_R	emp_R	res_R
FF	DW	0-5	64621	0.9899	0.9979	0.0080	0.0041	0.0011	-0.0029	0.0060	0.0010	-0.0050
FF	DW	5-10	77103	0.9907	0.9898	-0.0009	0.0037	0.0038	0.0001	0.0056	0.0064	0.0008
FF	DW	10-15	77966	0.9893	0.9852	-0.0040	0.0043	0.0056	0.0013	0.0064	0.0092	0.0028
FF	DW	15-20	62241	0.9878	0.9828	-0.0050	0.0049	0.0066	0.0016	0.0073	0.0107	0.0034
FF	DW	20-25	41520	0.9874	0.9878	0.0003	0.0051	0.0054	0.0003	0.0075	0.0069	-0.0006
FF	DW	25-30	20407	0.9847	0.9920	0.0073	0.0062	0.0054	-0.0008	0.0091	0.0026	-0.0065
FF	DW	30-35	4872	0.9695	0.9877	0.0182	0.0123	0.0070	-0.0053	0.0182	0.0053	-0.0129
FF	DW	35+	2867	0.9755	0.9885	0.0130	0.0099	0.0077	-0.0022	0.0146	0.0038	-0.0108
FF	SW	0-5	46709	0.9800	0.9822	0.0022	0.0187	0.0158	-0.0029	0.0013	0.0020	0.0007
FF	SW	5-10	97999	0.9760	0.9820	0.0060	0.0224	0.0145	-0.0079	0.0016	0.0035	0.0019
FF	SW	10-15	180163	0.9710	0.9697	-0.0013	0.0271	0.0269	-0.0002	0.0019	0.0034	0.0015
FF	SW	15-20	240344	0.9682	0.9663	-0.0019	0.0297	0.0310	0.0012	0.0021	0.0027	0.0006
FF	SW	20-25	255278	0.9682	0.9673	-0.0009	0.0298	0.0310	0.0012	0.0021	0.0017	-0.0003
FF	SW	25-30	234030	0.9709	0.9728	0.0019	0.0272	0.0261	-0.0011	0.0019	0.0011	-0.0008
FF	SW	30-35	191377	0.9761	0.9762	0.0002	0.0224	0.0228	0.0005	0.0016	0.0009	-0.0006
FF	SW	35+	410728	0.9840	0.9833	-0.0007	0.0150	0.0164	0.0014	0.0010	0.0003	-0.0007
RB	DW	0-5	18813	0.9927	0.9995	0.0068	0.0016	0.0000	-0.0016	0.0057	0.0005	-0.0052
RB	DW	5-10	16557	0.9905	0.9931	0.0027	0.0021	0.0012	-0.0009	0.0074	0.0057	-0.0018
RB	DW	10-15	14499	0.9888	0.9901	0.0013	0.0025	0.0026	0.0001	0.0088	0.0074	-0.0014
RB	DW	15-20	13106	0.9877	0.9760	-0.0117	0.0027	0.0043	0.0016	0.0096	0.0198	0.0101
RB	DW	20-25	8677	0.9831	0.9689	-0.0142	0.0037	0.0060	0.0023	0.0132	0.0251	0.0119
RB	DW	25-30	4218	0.9798	0.9742	-0.0057	0.0044	0.0088	0.0043	0.0157	0.0171	0.0013
RB	DW	30-35	1770	0.9762	0.9763	0.0001	0.0052	0.0045	-0.0007	0.0186	0.0192	0.0006
RB	DW	35+	976	0.9713	0.9703	-0.0010	0.0063	0.0113	0.0050	0.0224	0.0184	-0.0040
RB	SW	0-5	3350	0.9786	0.9937	0.0152	0.0201	0.0054	-0.0147	0.0013	0.0009	-0.0004
RB	SW	5-10	9999	0.9764	0.9878	0.0114	0.0222	0.0115	-0.0107	0.0014	0.0007	-0.0007
RB	SW	10-15	19093	0.9757	0.9863	0.0106	0.0228	0.0114	-0.0114	0.0015	0.0022	0.0008
RB	SW	15-20	26212	0.9788	0.9766	-0.0022	0.0199	0.0214	0.0015	0.0013	0.0020	0.0007
RB	SW	20-25	36841	0.9781	0.9716	-0.0065	0.0205	0.0269	0.0064	0.0013	0.0015	0.0002
RB	SW	25-30	37789	0.9740	0.9718	-0.0022	0.0244	0.0262	0.0018	0.0016	0.0020	0.0004
RB	SW	30-35	31888	0.9663	0.9674	0.0010	0.0316	0.0304	-0.0012	0.0021	0.0022	0.0002
RB	SW	35+	30722	0.9556	0.9567	0.0011	0.0417	0.0412	-0.0004	0.0027	0.0021	-0.0007

5.3 8-parameter + FE

Table 11: 8-parameter + FE per-cell fit. Same construction as Table 9.

Reg	Wall	Age	N	mod_M	emp_M	res_M	mod_E	emp_E	res_E	mod_R	emp_R	res_R
FF	DW	0-5	64621	0.9902	0.9979	0.0076	0.0040	0.0011	-0.0029	0.0058	0.0010	-0.0048
FF	DW	5-10	77103	0.9908	0.9898	-0.0010	0.0037	0.0038	0.0001	0.0055	0.0064	0.0009
FF	DW	10-15	77966	0.9890	0.9852	-0.0038	0.0044	0.0056	0.0011	0.0065	0.0092	0.0027
FF	DW	15-20	62241	0.9875	0.9828	-0.0047	0.0051	0.0066	0.0015	0.0074	0.0107	0.0032
FF	DW	20-25	41520	0.9874	0.9878	0.0004	0.0051	0.0054	0.0002	0.0075	0.0069	-0.0006
FF	DW	25-30	20407	0.9846	0.9920	0.0073	0.0062	0.0054	-0.0008	0.0091	0.0026	-0.0065
FF	DW	30-35	4872	0.9750	0.9877	0.0127	0.0101	0.0070	-0.0032	0.0148	0.0053	-0.0095
FF	DW	35+	2867	0.9840	0.9885	0.0045	0.0065	0.0077	0.0012	0.0095	0.0038	-0.0056
FF	SW	0-5	46709	0.9790	0.9822	0.0032	0.0197	0.0158	-0.0039	0.0014	0.0020	0.0007
FF	SW	5-10	97999	0.9759	0.9820	0.0062	0.0226	0.0145	-0.0081	0.0015	0.0035	0.0019
FF	SW	10-15	180163	0.9707	0.9697	-0.0010	0.0274	0.0269	-0.0005	0.0019	0.0034	0.0015
FF	SW	15-20	240344	0.9671	0.9663	-0.0008	0.0307	0.0310	0.0002	0.0021	0.0027	0.0006
FF	SW	20-25	255278	0.9677	0.9673	-0.0004	0.0302	0.0310	0.0008	0.0021	0.0017	-0.0003
FF	SW	25-30	234030	0.9715	0.9728	0.0013	0.0266	0.0261	-0.0006	0.0018	0.0011	-0.0007
FF	SW	30-35	191377	0.9769	0.9762	-0.0006	0.0217	0.0228	0.0012	0.0015	0.0009	-0.0006
FF	SW	35+	410728	0.9845	0.9833	-0.0012	0.0145	0.0164	0.0019	0.0010	0.0003	-0.0007
RB	DW	0-5	18813	0.9928	0.9995	0.0066	0.0015	0.0000	-0.0015	0.0056	0.0005	-0.0051
RB	DW	5-10	16557	0.9899	0.9931	0.0032	0.0021	0.0012	-0.0009	0.0079	0.0057	-0.0023
RB	DW	10-15	14499	0.9881	0.9901	0.0020	0.0025	0.0026	0.0000	0.0094	0.0074	-0.0020
RB	DW	15-20	13106	0.9872	0.9760	-0.0112	0.0027	0.0043	0.0015	0.0101	0.0198	0.0097
RB	DW	20-25	8677	0.9825	0.9689	-0.0136	0.0037	0.0060	0.0023	0.0138	0.0251	0.0113
RB	DW	25-30	4218	0.9797	0.9742	-0.0055	0.0043	0.0088	0.0045	0.0160	0.0171	0.0011
RB	DW	30-35	1770	0.9775	0.9763	-0.0013	0.0048	0.0045	-0.0003	0.0177	0.0192	0.0015
RB	DW	35+	976	0.9759	0.9703	-0.0056	0.0051	0.0113	0.0062	0.0190	0.0184	-0.0005
RB	SW	0-5	3350	0.9753	0.9937	0.0185	0.0231	0.0054	-0.0177	0.0016	0.0009	-0.0008
RB	SW	5-10	9999	0.9732	0.9878	0.0146	0.0250	0.0115	-0.0135	0.0018	0.0007	-0.0011
RB	SW	10-15	19093	0.9736	0.9863	0.0127	0.0246	0.0114	-0.0132	0.0018	0.0022	0.0005
RB	SW	15-20	26212	0.9782	0.9766	-0.0016	0.0204	0.0214	0.0011	0.0015	0.0020	0.0005
RB	SW	20-25	36841	0.9783	0.9716	-0.0067	0.0202	0.0269	0.0067	0.0014	0.0015	0.0000
RB	SW	25-30	37789	0.9742	0.9718	-0.0024	0.0240	0.0262	0.0022	0.0017	0.0020	0.0003
RB	SW	30-35	31888	0.9665	0.9674	0.0008	0.0312	0.0304	-0.0008	0.0022	0.0022	0.0000
RB	SW	35+	30722	0.9561	0.9567	0.0006	0.0410	0.0412	0.0003	0.0029	0.0021	-0.0009